

Library Management System - Capstone

Full Stack Development Project

User Manual

Overview

- LMS allows you to borrow books and make reservations of books not available
- You can also manage your book loans and reservations with the systems

LMS Membership

- ❑ Only members are allowed to borrow books, you need to create an account with your email to be a member
- ❑ Membership lasts for 1 year after account creation after which you need to requests for renewal.
- ❑ You cannot borrow any books if your membership is suspended or expired

Book Loan Rules

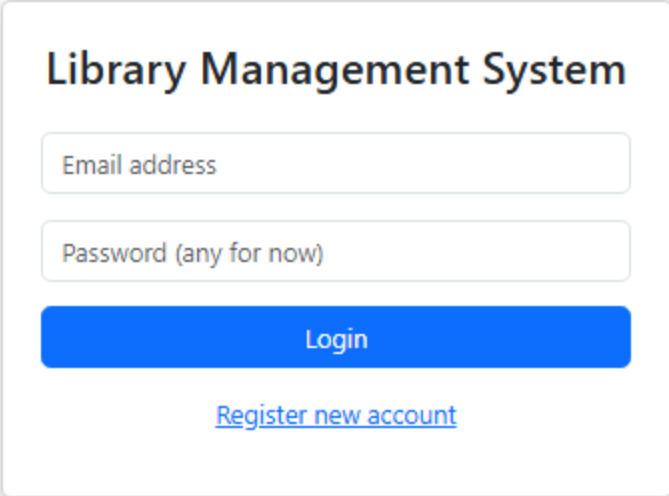
- You can loan up to 3 books at any one time, this will include any active loans that you have
- Date due for each book is 14 days upon checkout
- You can request for renew before date due is up for up to 2 time, each time the date due will move 14 days forward from date/time of renewal
- If a book is overdue for more than 1 day, a fine will be imposed at \$0.50 per day (up to a maximum of \$20 per book)
- If you have cumulative loans of \$10 or more, you cannot borrow any more books until you pay up the fines.
- You cannot borrow a book that has been reserved.

Book Reservation Rules

- You can reserve books already borrowed, up to 3 copies
- Reservation has a due date of 14 days
- If the book is not available by due date, reservation is automatically cancelled
- If the book is available and you did not borrow the book by date due, reservation is automatically cancel.
- You may cancel your reservation and release the book any time before due date.
- Same rules for maximum of 3 book loans apply.

Login to LMS

- Upon launch the web page, you will be presented by a login page
- You can login if you already have a valid account or you can register a new account.
- Click **register new account** if you are a new user



The image shows a login form for a 'Library Management System'. It features a title, two input fields for email and password, a blue login button, and a link to register a new account.

Library Management System

Email address

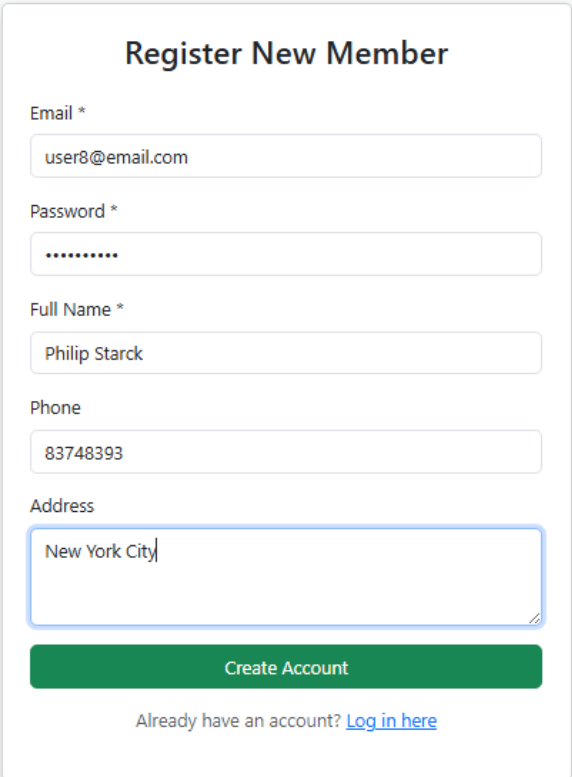
Password (any for now)

Login

[Register new account](#)

Register new account

- To register a new account, you must have an email address
- Enter your password and Full name
- Phone and Address is option
- Click Create Account and you will be notified when its successful
- Once your account is create, you will be brought back to the Login page for you to login to LMS



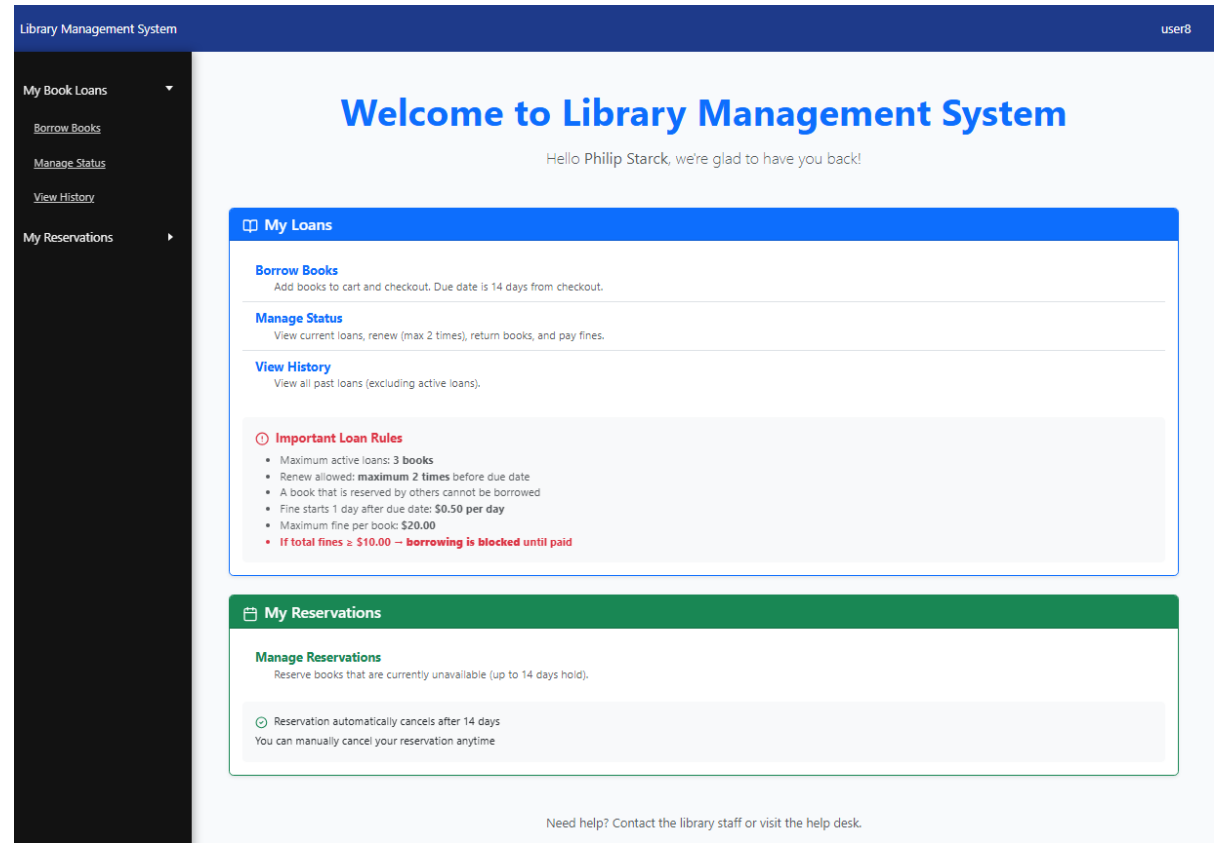
The image shows a 'Register New Member' form with the following fields and values:

- Email ***: user8@email.com
- Password ***: (masked)
- Full Name ***: Philip Starck
- Phone**: 83748393
- Address**: New York City

At the bottom of the form is a green button labeled 'Create Account'. Below the button, there is a link: 'Already have an account? [Log in here](#)'.

Member Dashboard

- After successful login, you will come to the member dashboard
- It will explain some of the rules for borrowing and reserving books
- It also explains what each menu does
- The main menu is on the left and on the top right hand corner is the user menu



User Menu

- The user menu has Profile and Logout as menu item.
- Logout will end your session and log you out.
- Profile allows you to edit your profile and change your password
- It will also show you your member ID, join date and expiration date
- You need to renew your membership before the expiration.
- If your member is expired, you are not allowed to login (cannot borrow books) until you renew your membership.

The screenshot displays a web application interface for a library system. At the top, a dark blue header bar contains the text "user8" on the right. Below this, a white dropdown menu is open, showing "Profile" and "Logout" options. The main content area has a light blue background with the text "ent System" in large blue letters. A blue sidebar is partially visible on the left. The primary focus is a "My Profile" card with a blue header and a white body. The header includes the title "My Profile" and an "Edit" button. The body contains a table with member details and a form for editing profile information.

Member ID	Date Joined	Expires On	Status
MEM-0027	10 Dec 2025	10 Dec 2026	Active

Full Name *	Philip Starck
Email	user8@email.com
Phone	83748393
Address	New York City
Account Type	Member

Borrow Books

- You can view the list of books to borrow and the status of your loans
- You can also search for books by ISBN, title, author, category, publisher and year published
- Books that have 0 available copies are not available to be borrow
- The top right hand corner is the **Book Cart** which allows you to checkout your new loans
- You can see from the picture that you have 2 books in cart and the cart shows (2)
- To checkout or edit your selection, click the Book Cart.
- You can borrow up to a maximum of 3 books, including active loans.

Borrow Books Book Cart ²

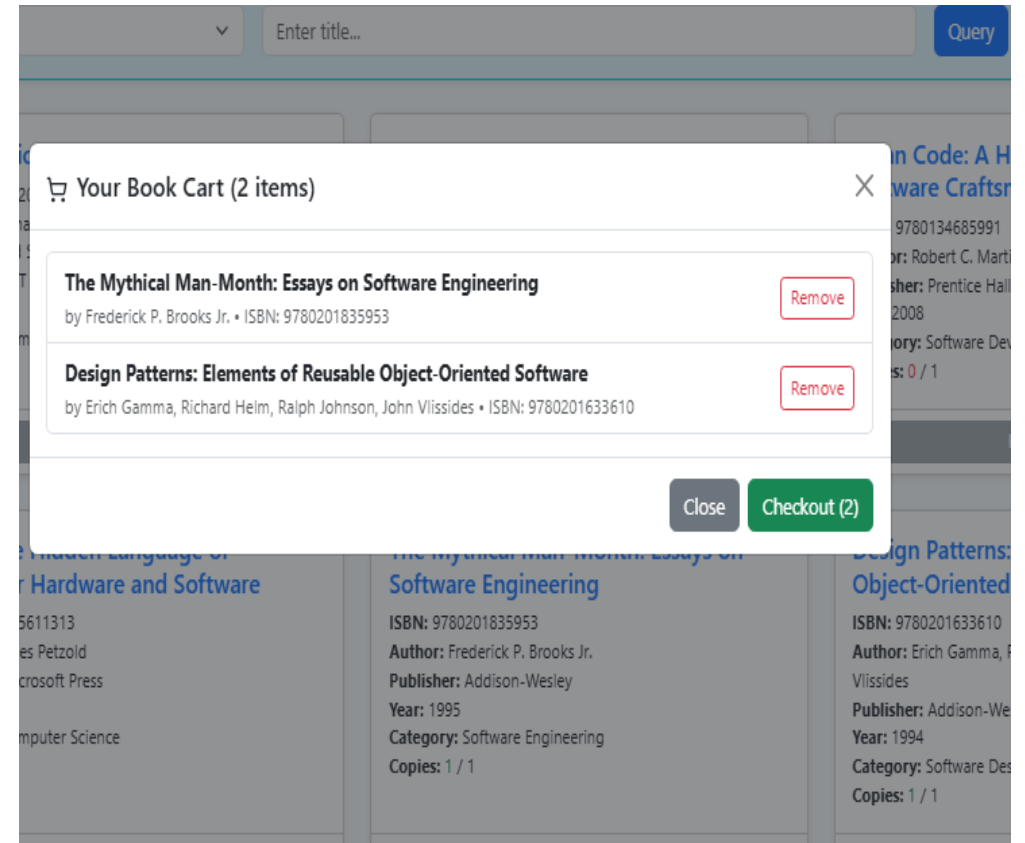
Currently Borrowed: 0 / 3 books — You can borrow 3 more Outstanding Fine: \$0.00

Search by: Title Search term: Enter title... Query Clear

Book Title	ISBN	Author	Publisher	Year	Category	Copies	Status
Introduction to Algorithms	9780262033848	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein	MIT Press	2009	Computer Science	0 / 1	Not Available
Structure and Interpretation of Computer Programs	9780262510875	Harold Abelson, Gerald Jay Sussman Jr.	MIT Press	1996	Programming	1 / 1	Add to Cart
Clean Code: A Handbook of Agile Software Craftsmanship	9780134685991	Robert C. Martin	Prentice Hall	2008	Software Development	0 / 1	Not Available
Code: The Hidden Language of Computer Hardware and Software	9780735611313	Charles Petzold	Microsoft Press	2000	Computer Science	1 / 1	Add to Cart
The Mythical Man-Month: Essays on Software Engineering	9780201835953	Frederick P. Brooks Jr.	Addison-Wesley	1995	Software Engineering	1 / 1	In Cart
Design Patterns: Elements of Reusable Object-Oriented Software	9780201633610	Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides	Addison-Wesley	1994	Software Design	1 / 1	In Cart

Borrow Books – Book Cart

- In the **Book Cart**, you can remove selections and checkout
- If you confirm that those are the books that you want to borrow, click checkout.
- The **Book Cart** will only remember your selection if you stay on the same page, once you click other pages, it will reset to 0.



Manage Status

- Here you can manage your book loans
- You can renew or return books from here
- You can renew up to a maximum of 2 more times for a book
- Fines are imposed on books past due, at \$0.5 per day, calculated one day from due date
- Each book can have fines up to a maximum of \$20.
- If the cumulative fines are \$10 or more, you cannot borrow or renew any books until you pay your fines.
- Click pay fines on the book to pay your fine. Paying a fine also returns the book automatically

My Current Loans

The Mythical Man-Month: Essays on Software Engineering

ID: 21 | Due: 24/12/2025 | Renewals: 0/2 | Fine: \$0.00

Renew

Return

Design Patterns: Elements of Reusable Object-Oriented Software

Returned just now

My Current Loans

Warning: You have unpaid fines. Please pay all fines before renewing any books. Paying a fine will also return the book.

1984

ID: 17 | Due: 04/12/2025 | Renewals: 0/2 | Fine: \$3.00

Renew

Return

Pay Fine & Return

Total Outstanding Fine: \$3.00

Pay All Fines & Return

View History

- This will show you the history of your book loans

My Loan History

 [Code: The Hidden Language of Computer Hardware and Software](#)

Borrowed: 04 Dec 2025 | Due: 04 Dec 2025

Returned

 [The Mythical Man-Month: Essays on Software Engineering](#)

Borrowed: 04 Dec 2025 | Due: 04 Dec 2025

Returned

 [The Art of Computer Programming, Volume 1: Fundamental Algorithms](#)

Borrowed: 04 Dec 2025 | Due: 04 Dec 2025

Returned

 [Atomic Habits](#)

Borrowed: 04 Dec 2025 | Due: 04 Dec 2025

Returned

 [Project Hail Mary](#)

Borrowed: 24 Dec 2025 | Due: 24 Dec 2025

Returned

 [The Fastest Zombie Alive](#)

Borrowed: 04 Dec 2025 | Due: 04 Dec 2025

Returned

Make Reservations

- Here you are reserved books that are on loans. You can reserve a maximum of 3 books
- Each reservation last for 14 days, no extension. After 14 days the reservation is expired.
- It works similar to **Book Loans** page.
- Instead of **Book Cart**, it's a **Reservation Cart**.
- You can checkout and remove selection from the cart similar to **Book Cart**

Make Reservations

Reservation Cart ¹

You have 0 active reservation(s)
You can reserve 3 more (Maximum 3)

Search By

Search Term

Title

Enter title...

Query

Clear

Introduction to Algorithms

ISBN: 9780262033848
Author: Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein
Publisher: MIT Press
Year: 2009
Category: Computer Science
Copies: 0 / 1

Reserve

Clean Code: A Handbook of Agile Software Craftsmanship

ISBN: 9780134685991
Author: Robert C. Martin
Publisher: Prentice Hall
Year: 2008
Category: Software Development
Copies: 0 / 1

Reserve

The Mythical Man-Month: Essays on Software Engineering

ISBN: 9780201835953
Author: Frederick P. Brooks Jr.
Publisher: Addison-Wesley
Year: 1995
Category: Software Engineering
Copies: 0 / 1

In Cart

Designing Data-Intensive Applications

ISBN: 9781449373320
Author: Martin Kleppmann
Publisher: O'Reilly Media
Year: 2017
Category: Databases
Copies: 0 / 1

The C Programming Language

ISBN: 9780131103627
Author: Brian W. Kernighan, Dennis M. Ritchie
Publisher: Prentice Hall
Year: 1988
Category: Programming
Copies: 0 / 1

Programming Pearls

ISBN: 9780201657883
Author: Jon Bentley
Publisher: Addison-Wesley
Year: 2000
Category: Programming
Copies: 0 / 1

Manage Reservation

- You can manage your reservations here
- If you cancel a reservations, it will be deleted.

Manage Reservations

[The Mythical Man-Month: Essays on Software Engineering](#)

Reserved: 10 Dec 2025 23:47 / Expires: 24 Dec 2025

X Cancel

Login notifications

- When you login it, the system checks for the following
 - If your status is Suspended, you are not allowed to log in; you must contact the Librarian
 - If your status is Expired, you can only login if you renew your membership
 - If your membership expiration date is 60 days or less, the system will ask you to renew your membership. If you choose to renew later, you will still be allowed to log in.
 - If you have any Loans are the past due date, you will get a notification upon logging into the system

END